

Practice Quiz: Systems

Name: __ Date: __

Part A: Multiple Choice

1. A system is best defined as: A) Any collection of parts B) A set of interacting parts that together produce behavior none of the parts produce alone — the whole has emergent properties C) Anything with a clear physical boundary D) A collection of parts that can be fully understood by studying each part individually
2. Which of the following is an open system? A) A sealed vacuum flask B) A living cell — it continuously exchanges matter and energy with its environment to survive C) A completely isolated crystal in space D) A closed chemical reaction in a sealed container
3. In a Markov Blanket, sensory states are: A) Internal states that the agent can control B) External states beyond the agent's reach C) States of the boundary through which the external world affects the agent — how outside information gets in D) The same as active states but in reverse
4. The key difference between homeostasis and allostasis is: A) Homeostasis is biological, allostasis is not B) Homeostasis reacts to current deviations; allostasis predicts and prepares for future deviations — allostasis is anticipatory C) Homeostasis uses energy, allostasis doesn't D) They are different words for the same process
5. A living system is "far from equilibrium" because: A) It is located in outer space B) It maintains a highly organized state that would decay without constant energy input — equilibrium = death for living things C) It is always moving and never stops D) It has more entropy than its environment
6. Which of these is the BEST example of allostasis? A) Your body sweating when overheated (reactive) B) A bear eating extra food before winter — anticipating future energy needs before the deficit occurs C) Your knee jerking when tapped (reflex) D) A thermostat adjusting temperature (reactive)

7. The Markov Blanket concept implies that: A) All systems are physically enclosed B) Internal and external states are conditionally independent given the blanket states — the inside "knows" the outside only through sensory and active states C) Systems cannot interact with their environment D) Boundaries are always rigid and fixed
8. Emergence means that: A) Things appear suddenly B) The system's properties cannot be inferred from the properties of its individual parts — the whole exhibits novel behavior C) Complex systems are always unpredictable D) Parts of a system can function independently

Part B: Short Answer

1. Your friend says: "A rock has a boundary (its surface), so it must have a Markov Blanket and be an agent." Is your friend correct? Explain why or why not. What does a rock have that an agent doesn't (or vice versa)? (200 words)
2. Consider a beehive. What are its sensory states and active states at the colony level? How is the hive's Markov Blanket different from the Markov Blanket of an individual bee? (200 words)

Part C: Essay

1. The human body maintains its temperature at approximately 37°C despite wildly varying environmental temperatures. Explain this using BOTH homeostasis (reactive mechanisms like sweating, shivering) and allostasis (predictive mechanisms like putting on a coat, moving to shade). Then explain how Active Inference unifies these two strategies under a single framework: minimizing the difference between predicted and actual internal states. (400 words)
2. Social media platforms (Facebook, TikTok, etc.) can be analyzed as systems. Define the Markov Blanket for a social media platform: What are its internal states? External states? Sensory states (how does information get in)? Active states (how does it affect the external world)? What does the platform "try" to maintain in its characteristic regime? Can a social media platform be considered an agent? Defend your answer. (400 words)